Work along a displacement

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate work and identify when a force does no work.

SUCCESS CRITERIA - I can...

1. Identify the force component along displacement.
2. Use meters and newtons to obtain joules.
3. Distinguish effort from mechanical work on an object.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “work” and “displacement” to explain your model or result. Sentence starters are optional.

work: energy transferred by a force along a displacement

displacement: change from initial position to final position, with direction

parallel: along the same direction as a line or surface

01 Warm-up: bring back an earlier idea

Keep impulse fixed, multiply target mass by 3 and lead time by 5.
Compare the modeled transverse shift using an equation.

Word help: lead time = time available before the modeled event

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

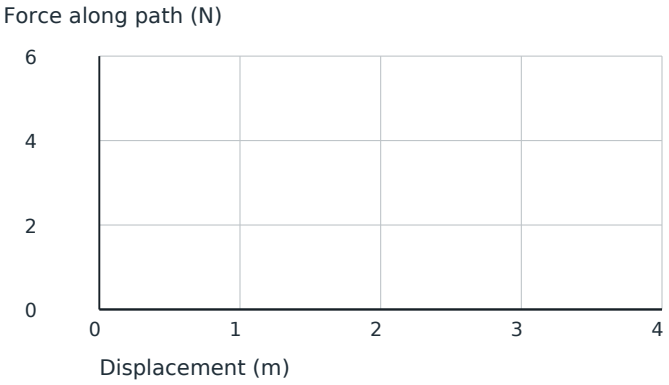
02 Worked example: explain the model

Reference: W = Fd.
Constant force parallel to displacement; W is work. No displacement means no work on the object by that force.

Worked reasoning: A constant 5 N pull acts along a 3 m displacement. Work is Fd=15 J. The object must move for this force to do mechanical work.

Your turn: Draw the force and displacement arrows.
Show the 15 J as area on a force-versus-position graph.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Work along a displacement

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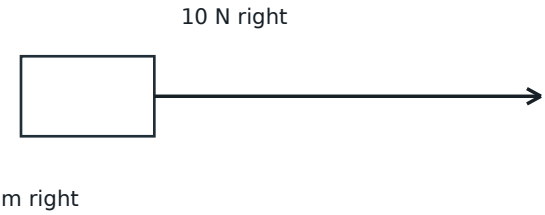
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A box moves 4 m right while a 10 N force acts right.
Draw force and displacement arrows; predict the sign of work.

Sentence starter: I predict … . My model supports this because … .



The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: 1 2 3

04 Apply the physics

A cart is pulled forward with a constant 18 N force over 5 m. A 6 N friction force opposes the motion.
Draw both forces and the displacement.
Calculate each force's work and their net work.

Sentence starter: I found … [unit]. This value means … in this situation.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student holds a heavy bag still. Is the upward support force doing mechanical work on the bag during the hold?
Explain without denying the student feels tired.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

Work along a displacement

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Continue down each column.

Sentence starters are optional. You may plan in any language.

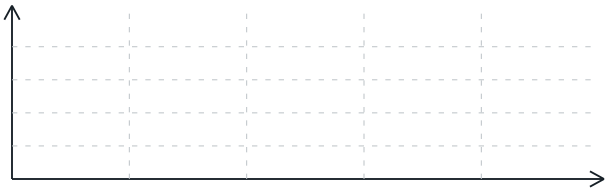
06 Review: connect an earlier idea

A 4 kg object slows from 8 to 4 m/s in 1.5 s.
Find impulse and average net force. Repeat the force calculation for the same velocity change over 3 s.
Show the signed impulse on the force-time model.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found ... [unit]. This value means ... in this situation.

Force (N)



Time (s)

Marzano check - circle: 1 2 3

07 Exit check

A force does 72 J over 3 m parallel to motion.
Find force. If displacement were zero, what work would that force do?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

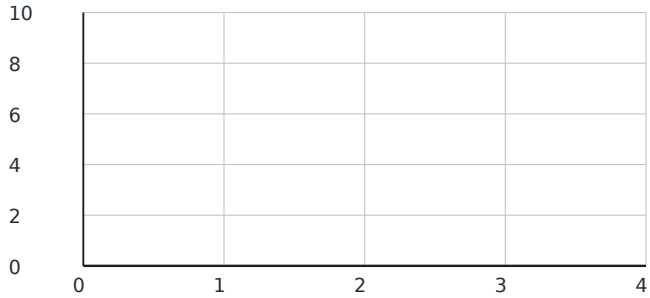
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A varying force rises linearly from 2 to 8 N over 4 m.
Find work from a graph and compare with using only the final force.

Sentence starter: Both ... . They differ in ... because ... .

Force along path (N)



Displacement (m)

Marzano check - circle: 1 2 3